Venkata Lakshmi Parimala Pasupuleti

Machine Learning Engineer · GenAI · Agentic AI · LLMOps · RAG Architectures · LangGraph · Python · LLM Evaluation · AWS

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SUMMARY

Machine Learning Engineer with 9+ years of production engineering depth and two years building GenAI and agentic AI systems end-to-end. Built **Leasel**: production RAG architecture with 8-agent LangGraph agentic workflows, context management, tool-use, multi-agent orchestration, HITL escalation, and responsible AI governance enforced in code - the full scope Cisco's CX AI platform requires. Built **AgentProbe**: open-source LLMOps toolkit with automated evaluation frameworks, model monitoring, CI regression harness, and LLM-as-judge scoring - keeps production AI quality measurable and reliable over time. Developed predictive models using scikit-learn, XGBoost, and fine-tuned transformer architectures (HuggingFace/PyTorch). At Entrata (9+ years, 100K+ users), owned production data pipelines, distributed systems, and AWS infrastructure in a regulated environment. Strong Python, Java, LangGraph/LangChain, Anthropic/OpenAI APIs, AWS, Docker, Kubernetes, CI/CD, and Claude Code (daily). M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

GenAI · Agentic AI Workflows | RAG Architectures · Hybrid Search | LangGraph · LangChain · LangSmith | LLMOps · Model Monitoring | Automated Evaluation Frameworks | Context Management · Embeddings | Multi-Agent Orchestration · HITL | Responsible AI Governance | Python · Java · FastAPI | scikit-learn · XGBoost · PyTorch · TensorFlow | Predictive Models · ETL Pipelines | Anthropic · OpenAI · Azure OpenAI | AWS (S3 · EC2 · RDS · Lambda) | Docker · Kubernetes · CI/CD | Low-Latency Model Serving | Claude Code · AI-Assisted Development | PostgreSQL · Redis · Vector Stores | Distributed Systems · REST APIs

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, data pipelines, and AI-powered application features; integrating AI components with backend APIs and surfacing insights into reporting layers.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built a production agentic workflow (LangGraph StateGraph) with RAG over regulatory documents, context management, validation guardrails, and conditional routing - natural-language inputs to validated structured outputs via low-latency FastAPI serving; deployed MCP server connecting agent outputs to enterprise systems.
- Engineered full document ingestion, chunking, retrieval, and streaming pipeline; applied responsible AI governance practices - hard compliance boundaries enforced before any export or action.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, multi-tenant)

- **Owned production data pipelines and ETL workflows** for 7+ years in a regulated environment - annual HUD compliance pipeline (few thousand federal rows → ~180,000 domain records) powering eligibility, rent, and compliance calculations for 100K+ housing units; zero compliance violations, zero security incidents; PHI-encrypted AWS S3 pipelines (Libsodium, 100% coverage).
- **Built and operated distributed event-driven processing** (RabbitMQ) with retry logic and idempotency; owned production incident response, root-cause analysis, and postmortems across distributed services; built internal debugging tools reducing debugging time by 40%.
- **Drove engineering excellence** through 300+ peer code reviews, automated test pipelines, and Python/Bash automation eliminating 80% of manual operational workload; led microservices redesign (Docker) - 50% scalability improvement, 30% deployment time reduction.
- Optimized 200+ SQL queries (30% gain) across complex multi-tenant PostgreSQL schemas; shipped 20+ full-stack production modules (Python, Java, PHP, JavaScript) across 120+ Agile sprints; mentored 5+ engineers.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built property management modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration; delivered large-scale data import workflows (1.6M records) under senior engineer guidance.

PROJECTS

Leasel - Production GenAI & Agentic AI Platform

2026 | Python, LangGraph (8-agent), FastAPI, RAG, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- **RAG architecture:** OCR-fallback ingestion, section-aware chunking, hybrid search (TF-IDF + semantic reranking), embedding-based retrieval, citation grounding to exact source clauses, context management balancing retrieval quality and token efficiency. **Agentic workflows:** 8-agent LangGraph StateGraph with durable state, tool-use, multi-agent orchestration (Triage → Retrieval → Configuration → Historical Incident → Reasoning → Validation → Human Escalation), and HITL escalation. **Responsible AI governance:** hard safety boundaries enforced in code - LLM reasons, deterministic or human logic makes every regulated compliance call; audit logging on every agent decision.
- 93.6% precision and 87.9% F1 across 1,240 test cases (10 compliance rules, 5 US states + federal).

AgentProbe - Open-Source LLMOps & Automated Evaluation Toolkit 2025 I Python, FastAPI, LangGraph, CI/CD I github.com/venkatalakshmiparimala/AgentProbe

- **LLMOps infrastructure:** automated evaluation frameworks with CI-integrated regression harness, model monitoring, LLM-as-judge scoring, and golden datasets - measures model quality, accuracy, latency, and hallucination rate across deployments. 15-endpoint FastAPI backend; multi-provider LLM routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama); used daily via Claude Code as an AI-assisted development tool to maintain production AI reliability.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017